
EE 201: Electromechanics

Course Code : EE 201

Course Name : Electromechanics

L-T-P-C : 2.5-0.5-0-3

Prerequisites : IC 160 Electrical System around Us

Students intended for : UG

Elective or Core : Core for EE, Elective for CSE/ME

Approval: 6th Senate

Course Contents

- **Circuits:** AC circuits - 1-phase & 3- phase (review); magnetic circuits. [2 Lectures]
- **Transformers:** 1-phase and 3-phase, auto-transformers, harmonics, special multiphase transformers and their applications. [8 Lectures]
- **Basic principles of electro mechanical energy conversion and rotating machines** [3 Lectures]
- **Synchronous Machines:** Construction, characteristics, regulation, V-curves, parallel operation and power system interfacing. [10 Lectures]
- **Induction machines:** 3-phase and 1- phase machine construction, characteristics, starting, braking and speed control, induction generators and applications. [10 Lectures]
- **Special machines:** Construction, characteristics and applications of a few special machines, such as, BLDC machine, PM machine, SRM, hysteresis motor, stepper motor, linear induction motor and their applications. [9 Lectures]

Text Books:

1. Kosow, I. L., **Electric Machinery & Transformers**, PHI, India
2. Nagrath I. J. and Kothari D. P., **Electrical Machines**, 3rd Edition, Tata McGraw-Hill Publishing Company Limited.
3. Venkatratnam K., **Special Electrical Machines**, University Press, India

Reference Books:

1. Fitzgerald A. E., Kingsley C. and Kusko A., **Electric Machinery**, 6th Edition, McGraw-Hill International Book Company.
2. Say M. G., **The Performance and Design of Alternating Current Machines**, CBS Publishers and Distributors.